CHAPTER 9 · DERIVATIONS

DEFORMATION OF SOLIDS

1
derivation the STPM 2026 syllabus asks you to derive, deduce or show

1 STRAIN ENERGY

9.3(B)

RESULT $U = \frac{1}{2}F\Delta l$

1. Up to the proportional limit, Hooke's law: $F = kx$.
2. Work done for a small extension dx: $dW = F\,dx = kx\,dx$.
3. Total work from 0 to Δl: $W = \int kx\,dx = \frac{1}{2}k(\Delta l)^2 = \frac{1}{2}(k\Delta l)\Delta l$.
4. **$U = \frac{1}{2}F\Delta l$** = area under the force–extension graph.

